Muhammad Haider Zaidi

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github.com/chefhaider Bavaria, Germany 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Bash, JavaScript
Tools/Frameworks: LangChain, LangSmith, Ragas, Hugging Face, FastAPI, Flask, TensorRT, PyTorch, TensorFlow, Docker, Kubernetes, AWS SageMaker, Streamlit, Plotly, Pandas, NumPy, scikit-learn, Pytest
Competence: LLM Agent Evaluation, Machine Learning, MLOps, Data Engineering, Experimental Design, Data Visualization, Model Benchmarking, Prompt Engineering, Statistical Analysis, ETL Pipelines

Experience

German National Metrology Institute (PTB) **March 2024 – Present**
Student Assistant Researcher *Berlin, DE*

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to enhance model accuracy and stability.
- Collaborated with PhD researchers to publish findings at the FLOMEKO conference.
- Developed dashboards for data visualization and model performance monitoring.

Folio3 **March 2022 – Nov 2023**
Machine Learning Engineer *California, US (remote)*

- Improved vitiligo segmentation accuracy to 94% and increased processing speed by 5x using AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and latency by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Conducted educational sessions on machine learning and big data at universities, enhancing team knowledge and collaboration.
- Designed POCs using YOLO, few-shot learning, and CNNs for client demonstrations.

Manda GmbH **March 2024 - June 2024**
Werkstudent Python Developer *Nürnberg, DE*

- Developed a full-stack website for AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and LLM tools for enhanced user interaction.

Freelance **July 2020 – March 2022**

Data Engineer *Remote*

- Built web scrapers and ETL pipelines for structured and unstructured data ingestion.
- Automated data cleaning and transformation workflows to support client analytics use cases.

Projects

CAFA 6: Protein Function Prediction **2022**

Deep Learning Competition

- Designed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Improved model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting **2022**

Thesis Project

- Developed a real-time Bitcoin price forecasting model using LSTM networks, achieving a 92% prediction accuracy.
- Implemented data preprocessing pipelines and visualized results using Plotly and Streamlit.

Saaz: AI Self-Playing Chitrali Sitar **2023**

Signal Processing, Music Generation

- Built the world's first AI-driven self-playing Chitrali Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI applications in music.

Hugging Face **Open-Source**

Contributor

- Co-authored Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.
- Contributed to improving LLM evaluation frameworks for enhanced model benchmarking.

Education

Friedrich Alexander Universität **Present**

Master of Science in Artificial Intelligence *Erlangen, DE*

- Focused on Machine Learning, LLMs, and Experimental Design for AI systems.

National University of Computer and Emerging Sciences **Aug 2018 – June 2022**

Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor for academic excellence.
- Specialized in Data Science and Machine Learning with a focus on real-world applications.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera**: TensorFlow Developer Certificate | **Databricks**: Lakehouse Fundamentals | **Datacamp**: Computer Vision in Python